FDA Submission

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Device: PneumoDetect-CXR

1. General Information

Intended Use Statement:

The device is intended to assist radiologists in the detection of pneumonia on frontal chest X-ray images by providing a binary classification (pneumonia present / not present) as a second-reader decision-support tool.

Indications for Use:

- Target population: Adult patients (18-90 years) undergoing frontal (PA or AP) chest radiography.

- Clinical setting: Screening or diagnostic chest X-ray workflows.

- Indicated only for images passing DICOM checks (Modality = DX, Body Part = CHEST, View = AP or PA).

- Not indicated for pediatric patients, lateral views, or patients with prior thoracic surgery.

Device Limitations:

- Not validated on non-frontal views, non-chest anatomy, or non-X-ray modalities.

- Training labels from NLP extraction of reports (accuracy greater than 90%).

- Pneumonia co-occurs with other findings; model may be influenced by correlated patterns.

Clinical Impact:

- False Negative: Missed pneumonia may delay treatment. Threshold favors sensitivity.

- False Positive: May lead to unnecessary antibiotics or additional workup.

2. Algorithm Design and Function

Algorithm Flowchart:

DICOM Input -> Header Checks (Modality=DX, BodyPart=CHEST, View=AP/PA)

| (reject if fail)

Pixel Array Extraction -> Preprocessing (resize 224x224, RGB conversion, rescale to [0,1])

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VGG16 Base (block5 fine-tuned) -> Dense Head (1024->512->256->1) -> Sigmoid Probability

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Threshold (0.523) -> Binary Output (Pneumonia / No Pneumonia)

Architecture:

Input (224x224x3) -> VGG16 Block 1-4 (frozen) -> VGG16 Block 5 (fine-tuned) -> Flatten -> Dense(1024) -> Dropout(0.5) -> Dense(512) -> Dropout(0.5) -> Dense(256) -> Dropout(0.3) -> Dense(1, Sigmoid)

DICOM Checking Steps:

- Modality must be DX

- BodyPartExamined must be CHEST or THORAX

- ViewPosition or PatientPosition must be AP or PA (warning logged if missing)

- Images failing checks are rejected

Preprocessing Steps:

- Convert grayscale to 3-channel RGB

- Resize to 224x224 (skimage.resize returns [0,1]; no additional /255)

- No mean/std standardization

3. Algorithm Training

Parameters:

- Augmentation: horizontal flip, rotation (+/-10), shift (+/-10%), shear (0.05), zoom (+/-10%)

- Batch size: 32

- Optimizer: Adam, learning rate 1e-4 with ReduceLROnPlateau

- Frozen: All VGG16 layers except block5

- Fine-tuned: block5 layers

Training Configuration Comparison:

Three configurations were evaluated:

- Original: LR 1e-4, 20 epochs, best val loss 0.494, AUC 0.684, F1 0.424

- Config A: LR 1e-4, 20 epochs, best val loss 0.455, AUC 0.683, F1 0.414

- Config C: LR 1e-4, ~30 epochs, best val loss 0.447, AUC 0.683, F1 0.416

All configurations performed similarly. Config C achieved the lowest validation loss. The final model uses Config C weights.

Training and Validation Loss Behavior:

Training loss decreased steadily while validation loss reached a minimum and then increased, indicating overfitting. The best checkpoint was retained. For Config C, best val loss was 0.447.

Final Threshold and Explanation:

The operating threshold of 0.523 was selected to maximize F1 on the validation set (F1 = 0.416). At this threshold: Sensitivity approximately 0.70, Specificity approximately 0.60, Accuracy approximately 0.62. This threshold balances sensitivity and specificity for a second-reader tool.

4. Databases

Training Dataset: NIH Chest X-ray Dataset (112,120 images from 30,805 patients). Labels from NLP extraction. Patient-level split balanced to approximately 50% pneumonia for training.

Validation Dataset: Held-out patient-level split with approximately 20% pneumonia prevalence. Size: 1,470 images. No augmentation applied.

5. Ground Truth

Labels generated by NLP of radiology reports (Wang et al., 2017). Accuracy estimated greater than 90%. Benefits: large-scale labeled dataset. Limitations: residual label noise; original reports not available for re-adjudication.

6. FDA Validation Plan

Patient Population: Adult patients (18-90) from multiple sites and scanner vendors. Balanced PA and AP views. Demographic diversity. Enrichment for pneumonia cases.

Ground Truth Methodology: Multi-reader multi-case (MRMC) with 3-5 radiologists. Majority vote or senior adjudicator. Clinical correlation where available. Report inter-reader agreement (kappa).

Performance Standard: Primary metrics: Sensitivity and specificity with 95% CI. Secondary: AUC, F1, precision-recall. Target: Sensitivity >= 0.80, Specificity >= 0.70, or non-inferior to radiologists. Literature: Radiologist F1 for pneumonia is 0.3-0.5 (Rajpurkar et al., CheXNet, 2017). Model F1 of 0.416 is within this range.